Machine Learning for Cross-Sectional Stock Return Prediction

Portfolio Construction & Backtesting Framework

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MATH 364 Final Project

Outline

Motivation

Why is return prediction hard?

- Financial markets are noisy and nonlinear
- Prices reflect firm characteristics, macro trends, and investor behavior
- Simple linear models miss important interactions

Our approach:

- Combine **technical**, **macroeconomic**, and **fundamental** features
- Compare linear vs. nonlinear ML models
- Evaluate economic value via a realistic backtest

Research Questions

1. Can ML models improve cross-sectional return prediction?
2. Do nonlinear models outperform the linear benchmark?
3. Are signals profitable after transaction costs?

Project Pipeline

Data → Features → Models → Prediction → Portfolio → Evaluation

Module 1 & 2 — Research

Data loading, feature construction, rolling ML prediction with semiannual hyperparameter tuning

Module 3 & 4 — Backtesting

Strategy backtest with signal smoothing, buffer-zone rules, and transaction cost adjustments

Data Sources

Stock Price Data

- U.S. large-cap equities
- Monthly adjusted closing prices
- 2016–2026 (monthly)

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1$$

Macro Data

- 3-month T-bill rate
- 10-year Treasury yield
- Consumer Price Index

Derived: **Term Spread** and **Inflation**

Fundamental Data

- PE, PB ratios
- ROE, Profit Margin
- Debt-to-Equity
- Market Cap

1-month lag to reduce look-ahead bias

Final Panel Dataset

Dataset summary:

- Shape: `(4294, 36)`
- Full range: Aug 2016 – Feb 2026
- Prediction range: Mar 2023 – Feb 2026
- Panel indexed by *stock* × *date*
- Observations with missing key variables removed

Table: Data Preview (selected columns)

Ticker	Ret_next	Mom1	Vol3
AAPL	$+0.066$	$+0.568$	$+0.528$
ADBE	$+0.061$	-0.416	-0.210
AMD	-0.066	$+3.445$	$+1.653$
AMZN	$+0.089$	$+0.141$	-0.006
BAC	-0.030	$+0.595$	$+1.252$

All values as of 2016-08-31

Feature Engineering Overview

Feature Group	Count	Examples
Technical	4	Mom1, Mom3, Mom6, Vol3
Macro	3	TB3MS, TermSpread, Inflation
Raw Fundamental	4	PE, PB, ROE, ProfitMargin
Ranked Fundamental	3	PE_Rank, ROE_Rank, MarketCap_Rank
Composite Fundamental	4	Value, Quality, Leverage, Overall
Seasonality	2	month_sin, month_cos

Key Feature Engineering Techniques

Cross-sectional Ranking

$$\text{Rank}_{i,t} = \frac{\text{rank}(x_{i,t})}{N_t}$$

Reduces extreme values; improves comparability across firms.

Composite Factors

- *Value*: PE + PB
- *Quality*: ROE + Profit Margin
- *Low Leverage*: D/E ratio
- *Fundamental*: weighted combination

1-Month Lag (Look-ahead Bias Reduction)

$$x_{i,t}^{\text{used}} = x_{i,t-1}$$

Seasonality Encoding

$$\text{month\_sin} = \sin\left(\frac{2\pi \cdot m}{12}\right)$$

$$\text{month\_cos} = \cos\left(\frac{2\pi \cdot m}{12}\right)$$

Captures cyclical calendar structure.

Three Models Compared

Ridge Regression

- Regularized linear model
- Traditional **benchmark**
- Captures linear relationships only

Random Forest

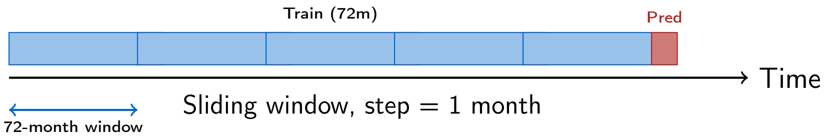
- Nonlinear ensemble
- Bagged decision trees
- Robust to overfitting
- Best performer

Gradient Boosting

- Sequential ensemble
- Captures complex patterns
- Higher turnover observed

Prediction Target: $\hat{R}_{i,t+1}$ — next-month cross-sectional return for each stock

Rolling Training Framework



Semiannual Hyperparameter Tuning

Hyperparameters tuned **every 6 months** — avoids overfitting to short-term noise while staying adaptive.

Strict Out-of-Sample Design

Each prediction uses only information available prior to the prediction date. No future data leakage.

Portfolio Construction

Signal Smoothing

- Predicted returns smoothed via **3-month moving average**
- Reduces short-term noise and excessive trading

Buffer-Zone Rule

- Enter only when ranking is **sufficiently extreme**
- Exit only when ranking falls below exit threshold
- Reduces turnover and improves stability

Baseline Parameters

Parameter	Value
Entry cutoff	Top/Bottom 15%
Exit cutoff	35% threshold
Rebalance freq.	Every 2 months
Smoothing window	3 months
One-way cost	10 bps

Net returns = gross returns − transaction costs

Main Strategy Results

Table: Baseline Strategy Performance (net of transaction costs)

Model	Ann. Ret.	Ann. Vol.	Sharpe	Final Val.	Max DD	Turnover	Hit Ratio
Random Forest	**23.1%**	22.9%	**1.007**	**1.785**	−16.9%	17.8%	61.8%
Gradient Boosting	6.9%	20.6%	0.337	1.150	−23.6%	23.8%	52.9%
Ridge Regression	−2.5%	26.8%	−0.093	0.843	−39.1%	19.8%	50.0%

Random Forest Highlights

- Highest Sharpe ratio: **1.007**
- Highest annualized net return: **23.1%**
- Lowest max drawdown: −16.9%
- Lowest turnover: 17.8%

Key Observation

Nonlinear models outperform the linear Ridge baseline. Technical, macro, and fundamental features contain useful **nonlinear interactions**.

Robustness Analysis

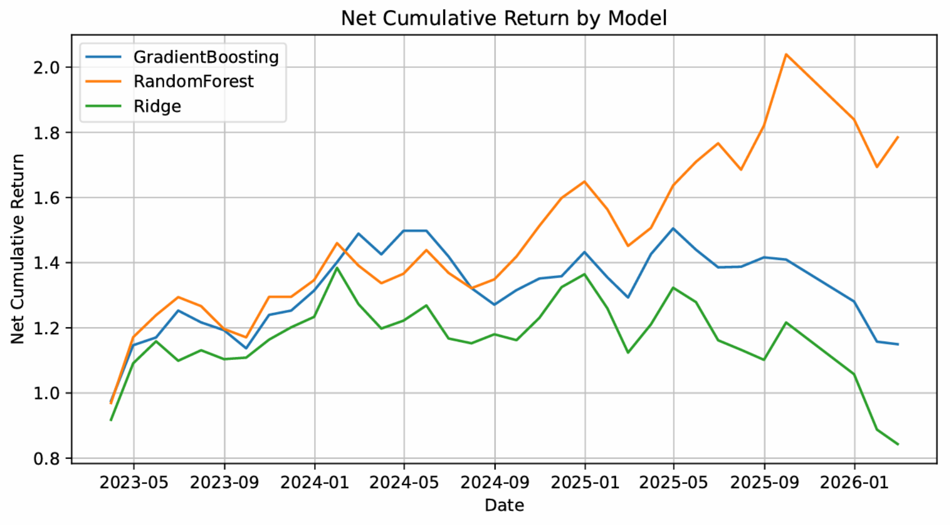
Table: Selected Specifications (ranked by Ann. Return)

Model	E / X / R / S	Ann. Ret.	Sharpe	Final Val.	Max DD
RF	0.10 / 0.25 / 1 / 1	**27.6%**	1.062	**1.981**	-15.6%
RF	0.15 / 0.35 / 2 / 3	23.1%	1.007	1.785	-16.9%
RF	0.20 / 0.45 / 3 / 3	19.5%	0.932	1.633	-16.5%
GB	0.20 / 0.45 / 3 / 3	19.3%	1.144	1.658	-12.9%
RF	0.20 / 0.40 / 2 / 3	13.9%	0.659	1.394	-18.3%
GB	0.20 / 0.40 / 2 / 3	9.5%	0.522	1.252	-19.3%

RF = Random Forest, GB = Gradient Boosting *E = entry, X = exit, R = rebalance freq., S = smoothing (months)*

Takeaway: Random Forest dominates across settings. The baseline balances return, turnover, and drawdown

Cumulative Return Plot



Key Insights

1. **Nonlinear models win.** Random Forest outperforms Ridge, confirming that nonlinear interactions among technical, macro, and fundamental features carry predictive value.
2. **Portfolio construction matters.** Signal smoothing + buffer-zone rules + lower rebalancing frequency help reduce unnecessary turnover and transaction costs, which improves the stability of the portfolio.
3. **Transaction costs are non-trivial.** Gradient Boosting's high turnover (23.8%) significantly erodes gross returns; net discipline is essential.
4. **Results are research evidence, not a trading system.** Short test period and non-point-in-time data require cautious interpretation.

Limitations & Future Work

Current Limitations

- Non point-in-time fundamentals (cached annual data)
- Limited stock universe
- Simplified transaction cost model (no liquidity or market impact)
- Short 36-month test period
- No industry neutralization

Future Improvements

- Institutional point-in-time databases
- Broader and more diverse universe
- Sector-neutral portfolio construction
- Realistic execution cost modeling
- Alternative architectures (LSTM, Transformer)
- Extended multi-regime testing

Conclusion

What we built:

- Complete ML pipeline: data $\rightarrow$ features $\rightarrow$ models $\rightarrow$ backtest
- Rolling 72-month training + semiannual HP tuning
- Buffer-zone portfolio with transaction cost modeling

Best result (Random Forest):

- Net Sharpe ratio $\approx$ **1.007**
- Annualized net return $\approx$ **23.1%**
- Final portfolio value: $\mathbf{1.785}\times$
- Maximum drawdown: only $-$**16.9%**

Summary

- Nonlinear ML > linear benchmark
- Structured features improve stability
- Disciplined construction controls costs
- Results suggest meaningful cross-sectional predictive signals

Thank You

Questions & Discussion

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